



---

Ministry of Higher Education and Scientific Research

University of Carthage

National Institute of Applied Science and Technology

Department of Physics and Instrumentation

Industrial Instrumentation and Maintenance

### **End-of-Year Project**

---

# **Smart Inspection of Solar Power Plants Using Drones and AI**

---

Prepared by

**Aziz Moualhi**

Date: 01/06/2026

Jury Members

**Mr Moncef Ben Smida : Supervisor**

**Mr Hosni Saidi : Examiner**

Academic Year : 2025/2026

# Acknowledgements

We would like to express our sincere gratitude to Mr. Moncef Ben Smida, our supervisor, for his valuable guidance, continuous support, and constructive advice throughout the duration of this project. His expertise and availability were essential to the successful completion of this work.

We also extend our thanks to Mr. Hosni Saidi for agreeing to examine this work and for his contribution as a jury member.

Finally, we wish to thank the teaching staff of the Department of Physics and Instrumentation at INSAT for the knowledge and skills they have transmitted throughout our academic journey.

# Abstract

The rapid expansion of photovoltaic (PV) installations has increased the need for efficient, scalable, and automated inspection methods to ensure system reliability and performance. Traditional manual inspection techniques are time-consuming, labor-intensive, and often insufficient for large-scale solar farms. To address these limitations, this project proposes an intelligent and automated inspection system for photovoltaic power plants based on drone-acquired imagery and artificial intelligence.

The proposed system integrates both RGB and thermal imaging to provide comprehensive defect detection and analysis. A dual-model deep learning approach is developed, consisting of specialized object detection models trained to identify different types of photovoltaic anomalies. In addition, a classification model is used to distinguish between image types, ensuring that each input is processed using the appropriate pipeline.

The system processes aerial data captured by drones and performs automated fault detection, severity assessment, and visualization of results through an interactive dashboard. This dashboard supports maintenance decision-making by presenting detected defects in a clear and interpretable format.

Experimental results show that the proposed approach improves inspection efficiency, reduces manual intervention, and enhances the accuracy and scalability of photovoltaic system monitoring. Overall, the system contributes to more reliable maintenance strategies and supports the transition toward smarter and more sustainable solar energy management.

# Contents

## General Introduction

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| <b>1</b> | <b>Theoretical Background</b>                               | <b>1</b>  |
| 1.1      | Introduction . . . . .                                      | 1         |
| 1.2      | Photovoltaic Systems Overview . . . . .                     | 1         |
| 1.2.1    | System Hierarchy: Cell, Module, String, and Array . . . . . | 1         |
| 1.2.2    | Photovoltaic Power Conversion Chain . . . . .               | 2         |
| 1.3      | Common PV Panel Defects . . . . .                           | 3         |
| 1.3.1    | Visible Surface Defects . . . . .                           | 3         |
| 1.3.2    | Thermal and Electrical Defects . . . . .                    | 4         |
| 1.4      | Inspection Methods . . . . .                                | 10        |
| 1.4.1    | Traditional Inspection Methods . . . . .                    | 10        |
| 1.4.2    | Drone-Based Inspection . . . . .                            | 12        |
| 1.5      | Sensing Technologies . . . . .                              | 12        |
| 1.5.1    | RGB Imaging . . . . .                                       | 12        |
| 1.5.2    | Thermal Imaging (Infrared Thermography) . . . . .           | 13        |
| 1.5.3    | Multi-Sensor Inspection Systems . . . . .                   | 14        |
| 1.6      | Artificial Intelligence in Inspection . . . . .             | 14        |
| 1.7      | Deep Learning for Object Detection . . . . .                | 14        |
| 1.7.1    | Convolutional Neural Networks (CNNs) . . . . .              | 14        |
| 1.7.2    | Object Detection Concepts . . . . .                         | 15        |
| 1.8      | Conclusion . . . . .                                        | 16        |
| <b>2</b> | <b>Methodology and System Design</b>                        | <b>18</b> |
| 2.1      | Introduction . . . . .                                      | 18        |
| 2.2      | Dual-Model Strategy . . . . .                               | 18        |
| 2.3      | Model Selection . . . . .                                   | 18        |
| 2.4      | Image Type Classification Model . . . . .                   | 19        |
| 2.5      | Data Acquisition For Detection Models . . . . .             | 19        |
| 2.5.1    | Thermal Dataset . . . . .                                   | 19        |
| 2.5.2    | RGB Dataset . . . . .                                       | 21        |
| 2.6      | Dashboard Design . . . . .                                  | 22        |
| 2.6.1    | System Architecture . . . . .                               | 22        |

|          |                                                   |           |
|----------|---------------------------------------------------|-----------|
| 2.6.2    | Fault Severity Classification . . . . .           | 22        |
| 2.6.3    | Maintenance Recommendations . . . . .             | 23        |
| 2.7      | Evaluation Metrics . . . . .                      | 23        |
| 2.7.1    | Intersection over Union . . . . .                 | 23        |
| 2.7.2    | Precision, Recall, and F1-Score . . . . .         | 24        |
| 2.7.3    | Mean Average Precision . . . . .                  | 25        |
| 2.7.4    | Accuracy . . . . .                                | 25        |
| 2.8      | Conclusion . . . . .                              | 25        |
| <b>3</b> | <b>Implementation</b>                             | <b>27</b> |
| 3.1      | Introduction . . . . .                            | 27        |
| 3.2      | Development Environment . . . . .                 | 27        |
| 3.3      | Model Training . . . . .                          | 27        |
| 3.3.1    | Image Type Classifier (ResNet18) . . . . .        | 27        |
| 3.3.2    | Thermal Defect Detection Model (YOLOv8) . . . . . | 28        |
| 3.3.3    | RGB Defect Detection Model (YOLOv11) . . . . .    | 28        |
| 3.4      | Training Challenges . . . . .                     | 29        |
| 3.5      | Testing and Validation . . . . .                  | 29        |
| 3.6      | Conclusion . . . . .                              | 29        |
| <b>4</b> | <b>Results and Discussion</b>                     | <b>31</b> |
| 4.1      | Introduction . . . . .                            | 31        |
| 4.2      | Quantitative Evaluation . . . . .                 | 31        |
| 4.2.1    | Thermal Detection Model (YOLOv8) . . . . .        | 31        |
| 4.2.2    | RGB Detection Model (YOLOv11) . . . . .           | 32        |
| 4.2.3    | Image Type Classifier (ResNet18) . . . . .        | 34        |
| 4.3      | Qualitative Results and Visual Analysis . . . . . | 34        |
| 4.3.1    | Interface Overview . . . . .                      | 34        |
| 4.3.2    | Thermal Inspection Results . . . . .              | 35        |
| 4.3.3    | RGB Inspection Results . . . . .                  | 36        |
| 4.4      | Performance Analysis . . . . .                    | 38        |
| 4.4.1    | Strengths of the Proposed System . . . . .        | 38        |
| 4.4.2    | Weaknesses of the Proposed System . . . . .       | 38        |
| 4.5      | Limitations . . . . .                             | 38        |
| 4.6      | Conclusion . . . . .                              | 39        |
| <b>5</b> | <b>Conclusion and Perspectives</b>                | <b>41</b> |
| 5.1      | Conclusion . . . . .                              | 41        |
| 5.2      | Future Work . . . . .                             | 41        |
|          | <b>Bibliography</b>                               | <b>43</b> |

# List of Figures

|      |                                                                                    |    |
|------|------------------------------------------------------------------------------------|----|
| 1.1  | PV system hierarchy: Cell → Module → String → Array. . . . .                       | 2  |
| 1.2  | Cracked photovoltaic panel. . . . .                                                | 3  |
| 1.3  | Examples of soiling on photovoltaic panels. . . . .                                | 4  |
| 1.4  | Thermal image showing a single hot spot on a photovoltaic panel. . . . .           | 5  |
| 1.5  | Thermal image showing multiple hot spots across a photovoltaic installation. . .   | 5  |
| 1.6  | Single-diode fault: visual pattern and thermal signature. . . . .                  | 6  |
| 1.7  | Multi-diode fault: visual pattern and thermal signature. . . . .                   | 7  |
| 1.8  | Bypass diode activation in a PV module under partial shading conditions. . . .     | 7  |
| 1.9  | Thermal images showing single-bypassed faults on photovoltaic panels. . . . .      | 8  |
| 1.10 | Double bypassed fault: visual pattern and thermal signature. . . . .               | 8  |
| 1.11 | String-open-circuit fault: visual pattern and thermal signature. . . . .           | 9  |
| 1.12 | String-reversed-polarity fault: visual pattern and thermal signature. . . . .      | 10 |
| 1.13 | Technician performing manual visual inspection of photovoltaic panels. . . . .     | 10 |
| 1.14 | Handheld thermal camera used for manual thermography inspection of PV panels.      | 11 |
| 1.15 | Current-Voltage curve. . . . .                                                     | 12 |
| 1.16 | Drone equipped with a thermal camera for inspection of PV installations. . . .     | 13 |
| 1.17 | Architecture of Convolutional Neural Networks. . . . .                             | 15 |
| 2.1  | Example of YOLO annotation format . . . . .                                        | 21 |
| 2.2  | Illustration of True Positive, False Positive, and False Negative detection cases. | 24 |
| 4.1  | Dashboard interface prior to image upload. . . . .                                 | 35 |
| 4.2  | Dashboard result for thermal image inspection. . . . .                             | 36 |
| 4.3  | Dashboard result for RGB image inspection. . . . .                                 | 37 |

# List of Tables

|     |                                                                     |    |
|-----|---------------------------------------------------------------------|----|
| 2.1 | Thermal dataset classes . . . . .                                   | 20 |
| 2.2 | RGB dataset classes . . . . .                                       | 21 |
| 2.3 | Defect severity classification for thermal and RGB models . . . . . | 22 |
| 2.4 | Maintenance actions for thermal defect classes . . . . .            | 23 |
| 2.5 | Maintenance actions for RGB defect classes . . . . .                | 23 |
| 3.1 | Image type classifier (ResNet18) – training configuration . . . . . | 28 |
| 3.2 | YOLOv8 thermal detection model – training configuration . . . . .   | 28 |
| 3.3 | YOLOv11 RGB detection model – training configuration . . . . .      | 28 |
| 4.1 | YOLOv8 thermal model – global performance metrics . . . . .         | 31 |
| 4.2 | YOLOv8 thermal model – per-class performance metrics . . . . .      | 32 |
| 4.3 | YOLOv11 RGB model – global performance metrics . . . . .            | 33 |
| 4.4 | YOLOv11 RGB model – per-class performance metrics . . . . .         | 33 |
| 4.5 | Image type classifier – validation performance . . . . .            | 34 |

# List of Abbreviations

| Abbreviation | Meaning                                   |
|--------------|-------------------------------------------|
| AI           | Artificial Intelligence                   |
| ANN          | Artificial Neural Network                 |
| CNN          | Convolutional Neural Network              |
| DC           | Direct Current                            |
| FN           | False Negative                            |
| FP           | False Positive                            |
| TP           | True Positive                             |
| IoU          | Intersection over Union                   |
| mAP          | Mean Average Precision                    |
| MPP          | Maximum Power Point                       |
| MPPT         | Maximum Power Point Tracking              |
| PV           | Photovoltaic                              |
| RGB          | Red Green Blue (visible-spectrum imaging) |
| UAV          | Unmanned Aerial Vehicle                   |
| YOLO         | You Only Look Once                        |

# General Introduction

## Context and Motivation

In recent decades, renewable energy has become a major global priority due to increasing energy demand, environmental concerns, and the need to reduce greenhouse gas emissions. Among renewable energy sources, solar photovoltaic (PV) technology has experienced remarkable growth because of its sustainability, accessibility, and decreasing installation costs. According to recent international reports, global photovoltaic capacity exceeded 2,260 GW in 2024, making solar energy one of the fastest-growing electricity sources worldwide [7].

The rapid deployment of photovoltaic power plants has led to the development of large-scale solar farms composed of thousands of interconnected modules operating under real environmental conditions. These installations are continuously exposed to factors such as dust, humidity, shading, temperature variations, and mechanical stress, which influence their long-term behavior and performance characteristics. Photovoltaic systems therefore represent complex energy infrastructures requiring continuous technical monitoring to ensure stable operation over time.

## Problem Statement

Traditional inspection of photovoltaic installations is mainly performed through manual on-site examination by technicians. This process becomes increasingly difficult as the size of solar farms grows, since it requires significant time, human effort, and operational cost. In addition, conventional visual inspection techniques are limited in their ability to identify certain types of defects, particularly those related to internal electrical behavior that may only appear as thermal variations. As a result, some anomalies remain undetected or are detected late, affecting system reliability and performance assessment. These limitations motivate the development of automated inspection approaches capable of analyzing photovoltaic installations in a more efficient and systematic way.

## Proposed Solution

This project proposes an intelligent inspection system for photovoltaic power plants based on drones and artificial intelligence techniques. The proposed approach uses unmanned aerial vehicles (UAVs), commonly known as drones, equipped with RGB and thermal cameras to capture aerial images of photovoltaic panels.

- RGB imaging allows the detection of visible defects such as cracks, dirt accumulation, broken cells, and physical degradation.
- Thermal imaging, on the other hand, enables the identification of abnormal temperature distributions associated with electrical defects and hot spots.

The collected images are processed using deep learning and computer vision algorithms. Two specialized artificial intelligence models are developed and trained for photovoltaic fault detection: The first model analyzes RGB images to detect visible anomalies, while the second model processes thermal images to identify temperature-related defects. By combining drone-based inspection with AI-driven image analysis, the proposed system aims to provide a faster, safer, and more efficient alternative to traditional photovoltaic inspection methods.

### Objectives of the Project

The main objective of this project is to develop an intelligent and automated inspection system for photovoltaic power plants using drones and artificial intelligence. The specific objectives are:

- To automatically detect faults and anomalies in photovoltaic panels.
- To use both RGB and thermal imaging for comprehensive defect analysis.
- To process aerial data acquired using drones.
- To design, train, and evaluate two specialized deep learning models for photovoltaic defect detection.
- To develop a visualization dashboard for real-time monitoring and interpretation of detected faults.
- To improve inspection speed and reduce dependence on manual maintenance operations.
- To assist maintenance teams through intelligent fault identification and analysis.
- To contribute to improving the efficiency, reliability, and sustainability of photovoltaic power plants.

### Report Organization

The report is organised into five chapters:

1. **Theoretical Background** – photovoltaic fundamentals, common defects, inspection methods, sensing technologies, and the fundamentals of artificial intelligence and deep learning for object detection.
2. **Methodology and System Design** – dual-model strategy, image type classification model, data acquisition process, dashboard design, fault severity classification, maintenance recommendations, and evaluation metrics.

3. **Implementation** – development environment, model training process for the image classifier and defect detection models (YOLOv8 and YOLOv11), training challenges, and validation procedures.
4. **Results and Discussion** – quantitative evaluation of the models, qualitative visual results, performance analysis, strengths, weaknesses, and system limitations.
5. **Conclusion and Perspectives** – summary of contributions and future improvements and research directions.

---

# Chapter 1

*Theoretical Background*

---

# Chapter 1

## Theoretical Background

### 1.1 Introduction

This chapter establishes the theoretical foundations of the project. It covers the principles of photovoltaic systems, common panel defects, available inspection methods, the sensing technologies used for data acquisition, and the artificial intelligence techniques applied to automated fault detection.

### 1.2 Photovoltaic Systems Overview

A photovoltaic (PV) system converts sunlight directly into electrical energy using semiconductor materials. The basic operating principle is the photovoltaic effect, where photons from sunlight generate electron movement inside solar cells, producing direct current (DC) electricity.

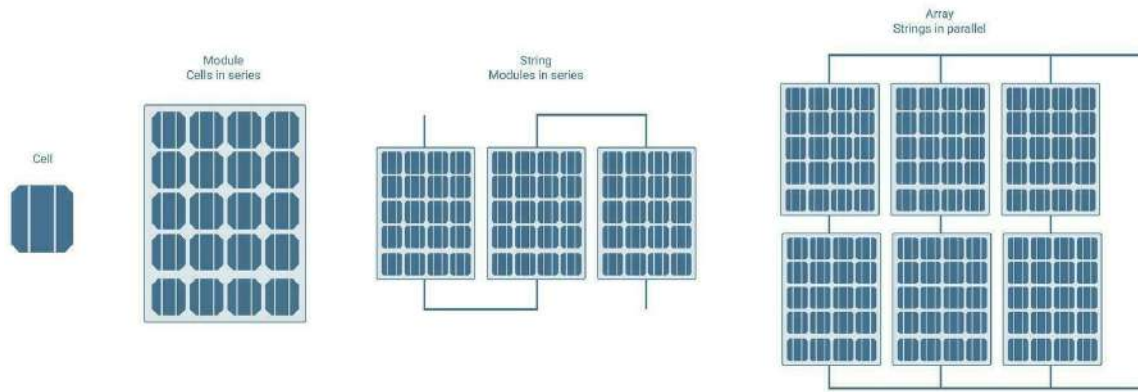
#### 1.2.1 System Hierarchy: Cell, Module, String, and Array

**Cell:** The cell is the basic unit responsible for energy conversion. Each cell is made of a semiconductor material, typically silicon, which generates electric current when exposed to sunlight. A single cell produces a very small amount of power ( $\sim 0.5$  W) and is therefore insufficient for practical applications.

**Module (panel):** A module is formed by connecting many solar cells in series and laminating them behind tempered glass. A typical module produces between 100 W and 500 W [8].

**String:** A string consists of multiple modules connected in series, which adds up their voltages. A string of 20 modules rated at 40 V each produces 800 V DC.

**Array:** An array is a combination of multiple PV strings connected in parallel. It represents the main energy-generating structure of a solar plant.



**Figure 1.1.** *PV system hierarchy: Cell → Module → String → Array.*

### 1.2.2 Photovoltaic Power Conversion Chain

**Combiner box:** The combiner box plays an essential role in merging parallel DC inputs into a single output. It also provides electrical protection through fuses and surge protection devices, ensuring safe operation in case of overcurrent or faults.

**MPPT and DC/DC converter:** DC power is processed by the maximum power point tracking (MPPT) system combined with a DC/DC converter.

- The MPPT is an intelligent control algorithm responsible for continuously adjusting the operating point of the photovoltaic modules in order to extract the maximum possible power under changing environmental conditions. Since the electrical characteristics of PV panels vary with sunlight and temperature, the MPPT ensures that the system always operates at its optimal point.
- The DC/DC converter adjusts voltage and current levels according to the MPPT control signals. It can either step up or step down the voltage to maintain stable operating conditions within the system.

**DC/AC inverter:** Once the DC power is optimized and stabilized, it is sent to the DC/AC inverter, which converts direct current into alternating current. This conversion is essential because electrical grids and most end users operate on AC power. The inverter also ensures grid synchronization by matching the frequency and phase of the produced electricity with that of the grid, enabling safe and efficient energy injection.

**Step-up transformer.** In large-scale plants, the AC output passes through a transformer to increase voltage, enabling efficient long-distance transmission.

**Electrical grid.** The converted electricity is injected into the distribution network for use by residential, industrial, and commercial consumers.

## 1.3 Common PV Panel Defects

### 1.3.1 Visible Surface Defects

#### Cracks

Cracks are fractures in PV modules that significantly impact performance and durability.

Their main causes include:

- **Manufacturing Defects:** Flaws during wafer production or module assembly can introduce weak points. Poor quality control may fail to detect such defects.
- **Mechanical Stress:** Improper handling during transportation or installation can lead to stress on the panels, causing cracks. Dropping or bending the panel increases the risk cracks can arise in a panel.
- **Environmental Stress:** Extreme weather conditions like hailstorms, strong winds (bending big panels), or temperature fluctuations introduce thermal expansion and contraction, which is leading to cracking.
- **Operational Wear:** Continuous exposure to environmental elements, combined with electrical load cycling, can cause micro cracks to propagate over time.



**Figure 1.2.** *Cracked photovoltaic panel.*

#### Soiling

Soiling refers to the accumulation of material on light-collecting surfaces, including dust, sand, pollen, soot, bird droppings, industrial pollutants, snow, frost, and biological films. Soiling reduces the amount of sunlight reaching the solar cells, thereby decreasing power output. The dominant type of soiling strongly depends on the geographic location and environmental conditions.

Two shading regimes are distinguished:

- **Hard shading:** Light is completely blocked. Affected cells stop producing electricity, causing current mismatch, potentially activating bypass diodes, and creating hotspots around shaded regions.
- **Soft shading:** Light is partially attenuated or scattered. Cells continue to generate electricity but with reduced efficiency and no complete shutdown.

The **soiling ratio (SR)** quantifies soiling impact, where  $SR = 1$  for a clean panel. SR is defined as:

$$SR = \frac{\text{Actual power output}}{\text{Expected power output if clean}} \quad (1.1)$$

The **soiling loss (SL)** represents the fraction of energy lost due to surface contamination and is defined as:

$$SL = 1 - SR \quad (1.2)$$



**Figure 1.3.** Examples of soiling on photovoltaic panels.

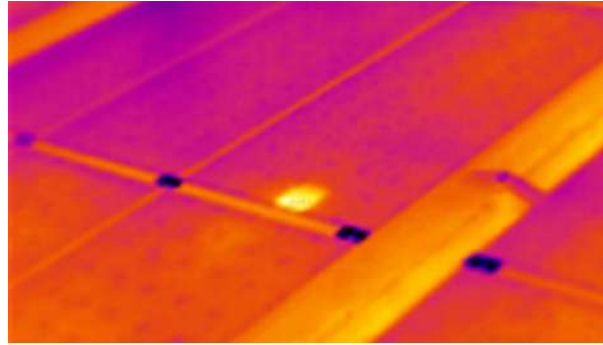
### 1.3.2 Thermal and Electrical Defects

#### Single HotSpot

A Single HotSpot is a localized area of high temperature on a single solar cell. it dissipates energy as heat instead of producing electricity

- **Cause:** Usually triggered by micro-cracks, shadows (bird droppings, leaves), or manufacturing defects in a specific cell.
- **Thermal Signature:** Appears as a bright, concentrated point of heat.

- **Impact:** Early-stage fault, If ignored, can evolve into Multi HotSpot or permanent damage

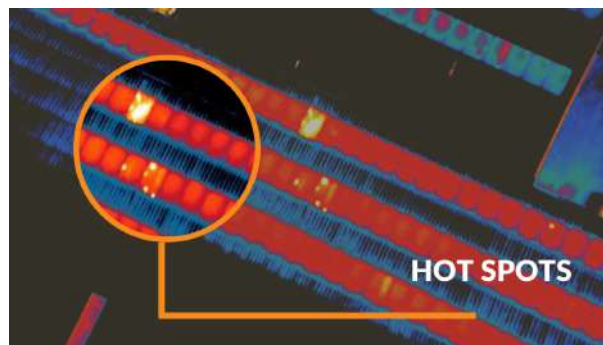


**Figure 1.4.** Thermal image showing a single hot spot on a photovoltaic panel.

### Multi Hotspot

This refers to multiple individual hotspots scattered across different cells or different panels within a string.

- **Cause:** Often indicates a widespread issue like consistent shading ( a nearby tree or fence) or a batch of low-quality cells experiencing "mismatch" losses.
- **Thermal Signature:** Multiple bright spots appearing simultaneously across the module.
- **Impact:** Accelerates aging of cells, can lead to burn marks or cracks, reduces panel lifespan.



**Figure 1.5.** Thermal image showing multiple hot spots across a photovoltaic installation.

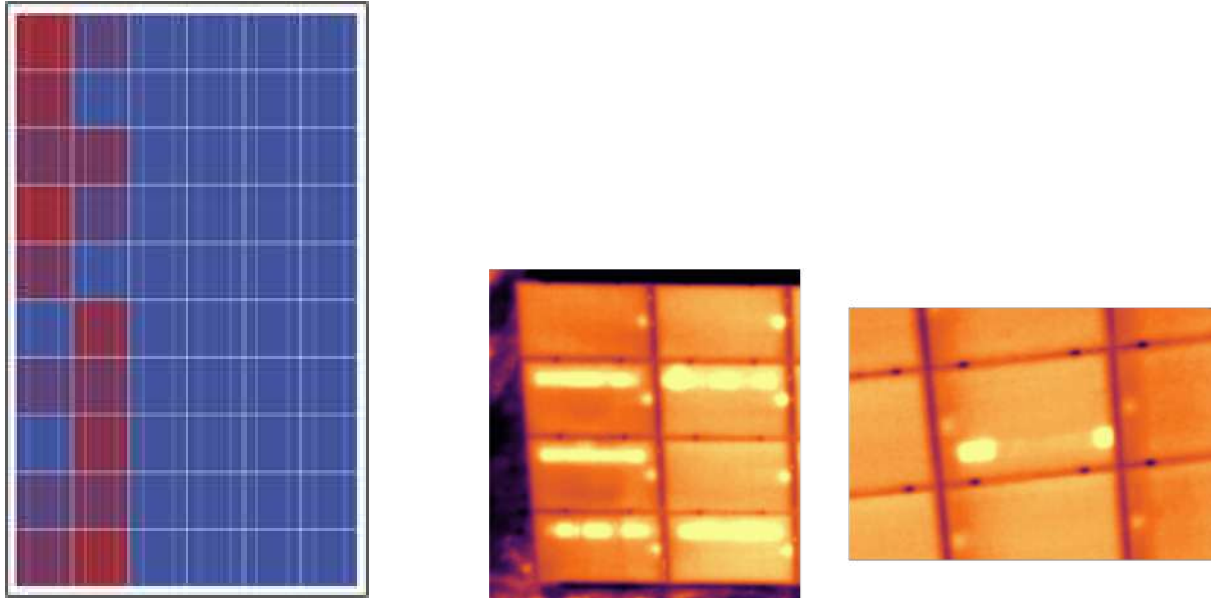
### Single Diode

A Single Diode defect occurs when a bypass diode associated with one substring of a photovoltaic module becomes defective, affecting the electrical behavior of that section.

- **Cause:** It is typically caused by electrical stress, aging, or manufacturing defects in the bypass diode.
- **Thermal Signature:** When several cells in a PV module are at a high temperature, they are known as Multi-hot Spots. However, if these cells are located in the same substring, the root cause will likely be the diode.

In such cases, one-third of the PV module typically shows a high-temperature variation in the cells, heating becomes non-uniform, while the rest has a consistent temperature. This issue is known as a single-diode problem.

- **Impact:** Uneven current distribution, reduced power output, potential long-term degradation of the affected substring.



(a) Single-diode fault pattern.

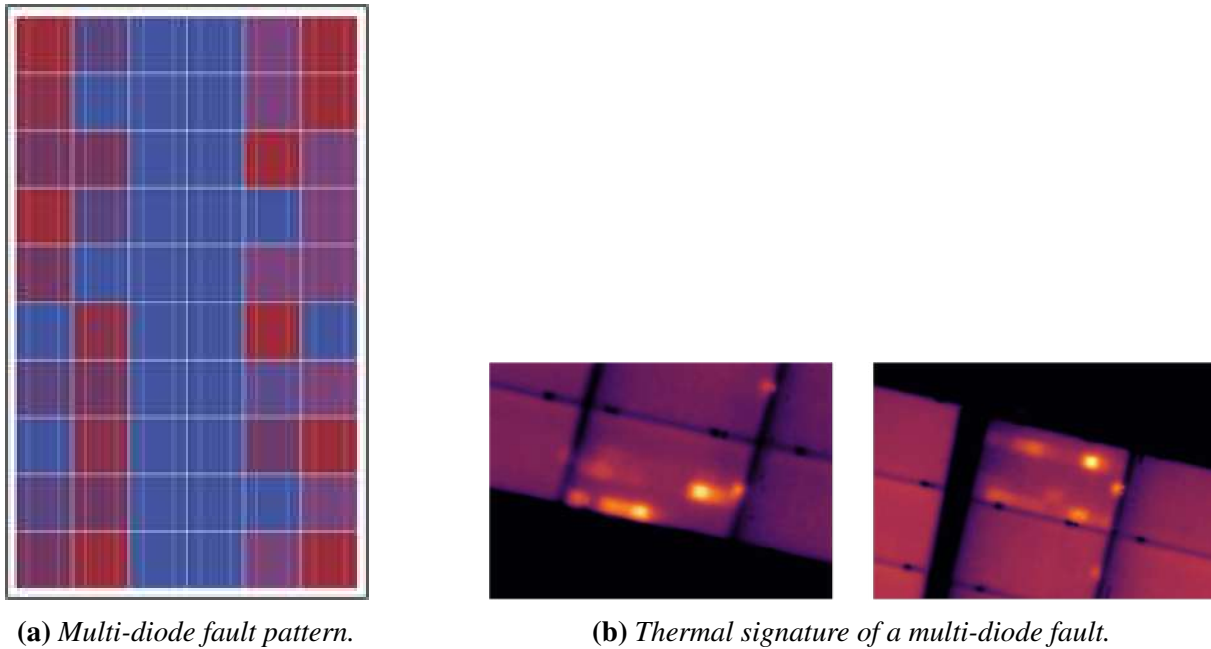
(b) Thermal signature of a single-diode fault.

**Figure 1.6.** Single-diode fault: visual pattern and thermal signature.

### Multi Diode

A Multi Diode defect occurs when multiple bypass diodes across different substrings of a photovoltaic module are simultaneously affected.

- **Cause:** It is usually caused by extended electrical stress, aging of multiple diodes, or severe operational conditions affecting several module sections.
- **Thermal Signature:** Thermal images show multiple affected substrings with non-uniform heating patterns. The difference compared to a multi-hotspot issue is that the anomalies follow the line of the substrings and appear linear
- **Impact:** Significant power loss, Reduced module efficiency, Increased risk of progressive system degradation.

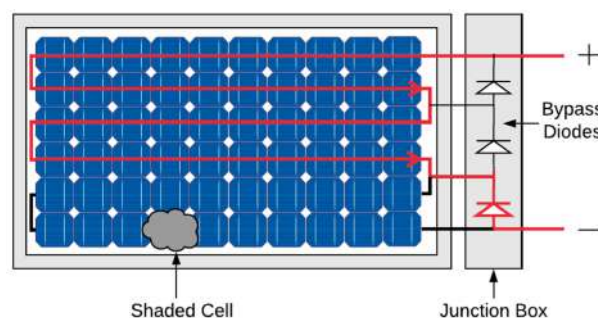


**Figure 1.7.** Multi-diode fault: visual pattern and thermal signature.

### Single Bypassed

A Single Bypassed condition occurs when one bypass diode is activated within a photovoltaic module.

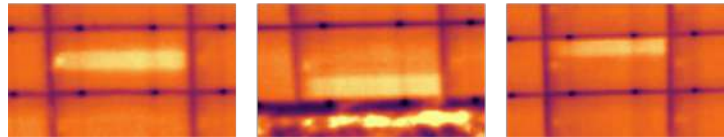
- **Cause:** The bypass diode is placed in parallel to sections of solar cells providing an alternate path for electrical current in the event of a partial shading of a panel or a malfunction. The activation of a bypass diode is a normal protective response when: A small portion of the panel is shaded, One section becomes faulty



**Figure 1.8.** Bypass diode activation in a PV module under partial shading conditions.

- **Thermal Signature:** The substrings that are isolated won't produce any energy. As a result, they will have a higher temperature than the cells working normally in the same PV module. The affected substring appears thermally uniform, indicating that current is being rerouted through the bypass diode.
- **Impact:** Minor power loss, not necessarily dangerous, but indicates a localized issue that should be monitored.

- **Common misconception:** It is common for people to mistake this defect with a single or multi-diode issue  
 single or double bypassed substring: The affected cells have the same temperature, the whole substring shows homogeneously hotter than other substrings in the same module. This thermal pattern shows that the bypass diode works properly  
 single or multi-diode issue: the thermographic images will display an irregular checkerboard pattern on the affected substring instead of a homogenous temperature increase.

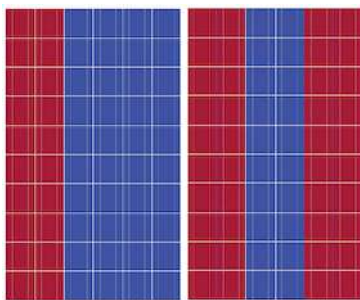


**Figure 1.9.** Thermal images showing single-bypassed faults on photovoltaic panels.

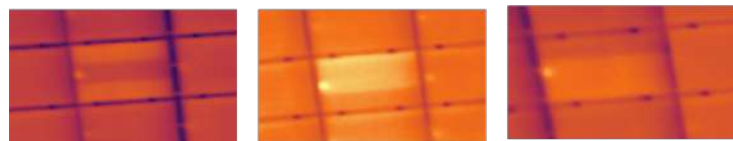
### Multi Bypassed

This defect refers to a situation where multiple bypass diodes are activated simultaneously in a photovoltaic module.

- **Cause:** It is caused by widespread shading, multiple faulty substrings, or significant mismatch conditions within the panel.
- **Thermal Signature:** Several substrings appear thermally distinct, often showing uniform temperature regions separated from normal operating areas.
- **Impact:** Significant loss of power output



**(a)** Single vs multi bypassed.



**(b)** Thermal signature of a double-bypassed fault.

**Figure 1.10.** Double bypassed fault: visual pattern and thermal signature.

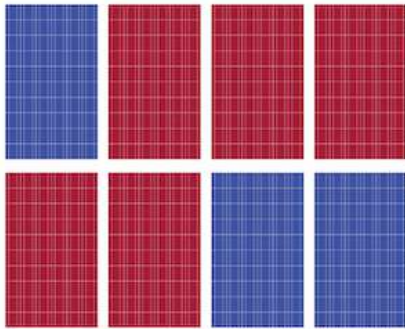
### String Open Circuit

A String Open Circuit refers to a condition where an entire photovoltaic string is electrically disconnected, preventing current flow through the affected modules.

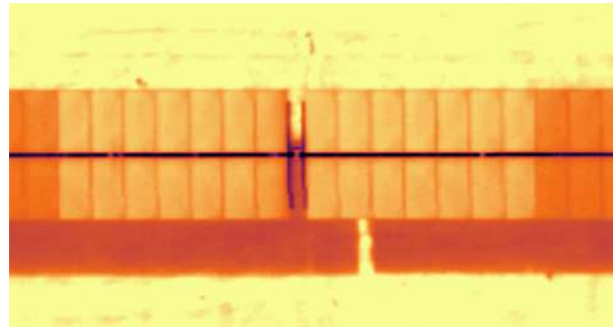
- **Cause:** a blown fuse, a disconnected cable, a damaged or burnt welding point, a physical issue in a connection box or at the inverter, any issue that could cause a physical disconnection

or prevent current from flowing through the string.

- **Thermal Signature:** One group of PV modules is warmer than another comparable group of PV modules by a few degrees since the solar irradiance isn't converted into power, but the temperature is uniform across the impacted group.
- **Impact:** Power Losses: when an entire string of PV modules is disconnected, these PV modules are not producing any power. This means that the power loss is 100% for as long as the issue has existed



(a) string-open-circuit fault pattern.



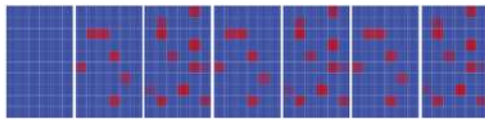
(b) Thermal signature of a string-open-circuit fault.

**Figure 1.11.** String-open-circuit fault: visual pattern and thermal signature.

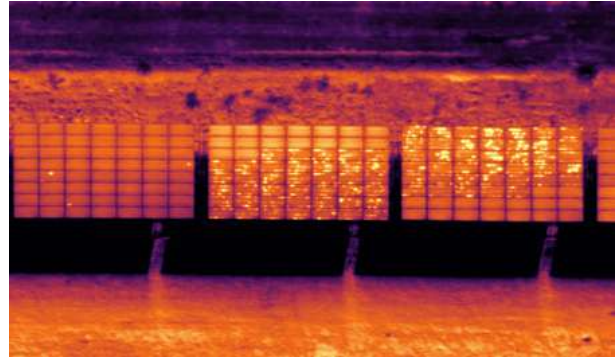
### String Reversed Polarity

A string reversed polarity fault occurs when the current through a string of PV panels is flowing in the wrong direction relative to normal operating conditions.

- **Cause:** The exact cause of this anomaly can be difficult to determine without on-site electrical measurements. It is generally associated with: short-circuit, low-voltage, reverse current flow, mismatch in string's working conditions.
- **Thermal Signature:** One group of contiguous panels shows a patchwork pattern (panels with multi-hotspots), usually the whole string.
- **Impact:** Severe performance degradation: the PV modules are not producing any power. the affected panels dissipate energy as heat rather than produce power. This means that the power loss is 100% for as long as the string issue has existed, Possible damage to the solar equipment.



(a) *String-reversed-polarity fault pattern.*



(b) *Thermal signature of String-reversed-polarity.*

**Figure 1.12.** *String-reversed-polarity fault: visual pattern and thermal signature.*

## 1.4 Inspection Methods

The performance and reliability of photovoltaic installations depend heavily on regular inspection and maintenance.

Various inspection techniques are used to identify defects, monitor degradation, and ensure optimal energy production.

These methods can be divided into traditional inspection and modern inspection systems.

### 1.4.1 Traditional Inspection Methods

#### Visual Inspection

Visual inspection is an effective and simple method for identifying failures such as burnt cells, cracks, delamination, and dust accumulation. This method is also inexpensive since it does not require sophisticated equipment. However, it is time-consuming, prone to human error, limited in scalability for large PV plants, and incapable of detecting internal electrical defects.



**Figure 1.13.** *Technician performing manual visual inspection of photovoltaic panels.*

#### Thermography Inspection

Infrared thermography is the most widely implemented technique in PV power plants. It excels at detecting mismatched cells, shaded cells, and defective bypass diodes. However, it's still

traditionally manual. Similar to on-the-ground visual inspections, technicians must visit each panel individually, interpret thermal results, and suggest corrective actions. These manual processes are error-prone and time-intensive and aren't the best option, with modern technology enabling new options.



**Figure 1.14.** Handheld thermal camera used for manual thermography inspection of PV panels.

### Current-Voltage (I–V) Measurements

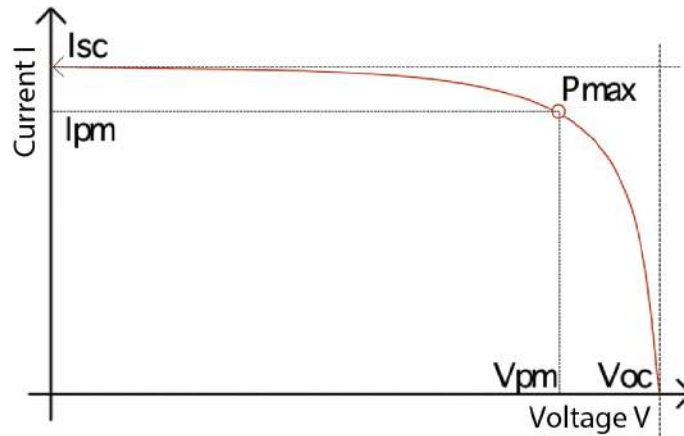
I–V measurements evaluate the electrical performance of solar cells by generating the characteristic curve relating voltage to current at specific irradiance and temperature conditions. Key parameters derived include:

- **Short-circuit current** ( $I_{sc}$ ): maximum current at zero voltage.
- **Open-circuit voltage** ( $V_{oc}$ ): maximum voltage at zero current.
- **Maximum Power Point (MPP)**:  $P_{max} = V_{mp} \times I_{mp}$ .
- **Fill factor (FF)**: defined as:

$$FF = \frac{V_{mp} \times I_{mp}}{V_{oc} \times I_{sc}} \quad (1.3)$$

A good panel typically has a fill factor between 0.75 and 0.85.

The measured I–V curve is compared with the manufacturer's reference characteristics to identify degradation, power losses, and electrical faults. Although this method provides precise electrical analysis, it is invasive (often requires disconnecting the modules from the PV system), slow, and difficult to scale for large installations.



**Figure 1.15.** *Current-Voltage curve.*

### Limitations of Traditional Methods

Although traditional inspection methods provide valuable information, they suffer from several common limitations: high dependence on human intervention, time-consuming procedures, limited scalability for large PV plants, risk of human error, invasive measurements in certain techniques, and need for specialised equipment and expertise.

#### 1.4.2 Drone-Based Inspection

Unmanned Aerial Vehicles (UAVs) have become an efficient solution for photovoltaic inspection. Drones equipped with cameras can autonomously fly over solar farms and capture high-resolution images for defect analysis. Modern solutions integrate drone-acquired data with inspection software for analysis, report generation, and predictive maintenance scheduling.

## 1.5 Sensing Technologies

### 1.5.1 RGB Imaging

RGB imaging uses conventional visible-light cameras to capture images in a manner similar to human vision. These cameras provide detailed visual information about the surface condition of photovoltaic panels and detect dust and soiling, cracks, broken glass, discoloration, shading, and physical damage.

**Advantages:** inexpensive, high spatial resolution, easy deployment.

**Limitations:** cannot detect internal electrical faults, affected by lighting conditions.

### 1.5.2 Thermal Imaging (Infrared Thermography)

Thermal imaging measures the infrared radiation emitted by photovoltaic modules to visualise surface temperature distribution. Electrical defects generate abnormal heat patterns detectable even in the absence of visible surface damage.

#### How a Thermal Camera Works:

- **Detection of infrared radiation:** All objects emit infrared (IR) radiation depending on their temperature. The warmer the object, the more IR energy it emits.
- **Capture of infrared energy:** The thermal camera has a special sensor (microbolometer). This sensor detects infrared radiation
- **Conversion into electrical signals:** The infrared energy is converted into electrical signals. Each pixel in the sensor corresponds to a temperature measurement.
- **Temperature mapping:** The camera assigns temperature values to each pixel. This creates a “temperature matrix” of the scene.
- **Image formation :** The temperature matrix is converted into a visual image. Different temperatures are represented by colors

These cameras help detect hotspots, defective cells, bypass diode failures, connection issues, electrical mismatches.

**Advantages:** detects hidden electrical faults, suitable for large-scale inspection.

**Limitations:** affected by weather conditions, lower image resolution than RGB, requires calibration and interpretation expertise, requires irradiance typically exceeding 600–700 W/m<sup>2</sup> for reliable fault visualisation.



**Figure 1.16.** Drone equipped with a thermal camera for inspection of PV installations.

### 1.5.3 Multi-Sensor Inspection Systems

Modern photovoltaic inspection systems increasingly rely on multi-sensor approaches combining RGB and thermal imaging. Integrating these complementary sensing modalities improves defect detection accuracy and system reliability.

## 1.6 Artificial Intelligence in Inspection

Artificial intelligence has become a key enabling technology in modern industrial inspection. Machine learning and deep learning techniques allow AI systems to automatically analyse images to detect, classify, and localise photovoltaic defects with high speed and accuracy.

AI-based inspection transforms raw image data into actionable information by learning patterns associated with common PV anomalies from large annotated datasets. This supports the transition from reactive maintenance—where faults are addressed after failure occurs—to predictive maintenance, where potential issues are identified before they lead to significant performance losses.

## 1.7 Deep Learning for Object Detection

### 1.7.1 Convolutional Neural Networks (CNNs)

CNNs are a type of artificial neural network widely used in deep learning and machine learning applications.

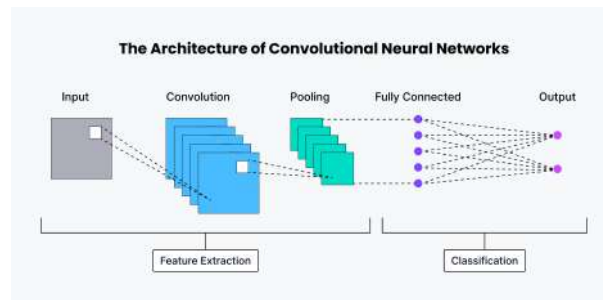
CNNs are inspired by the functioning of the human visual system and are designed to recognize patterns such as edges, textures, and shapes which makes them especially effective for processing image data and extracting important features automatically.

CNNs are commonly used in computer vision, image classification, facial recognition, medical imaging, and intelligent control systems.

#### Architecture of CNNs

- **Input Layer :** The input layer receives the image or data to be analyzed.
- **Convolutional Layer:** This layer applies filters (kernels) to extract important features such as edges, shapes, and textures.
- **Activation Function :** An activation function, usually ReLU, introduces non-linearity and improves the learning capability of the network.
- **Pooling Layer :** The pooling layer reduces the size of the feature maps, decreases computation time, and helps prevent overfitting.
- **Fully Connected Layer:** This layer uses the extracted features to perform the final classifi-

cation or prediction.



**Figure 1.17.** *Architecture of Convolutional Neural Networks.*

### Advantages of CNNs:

CNNs offer several important advantages:

- Automatic feature extraction
- High accuracy in image recognition tasks
- Reduced need for manual preprocessing
- Efficient handling of large datasets
- Ability to learn complex patterns.

### 1.7.2 Object Detection Concepts

Object detection in computer vision involves two simultaneous tasks:

- **Object Classification:** determining what objects are present in the image (e.g., cracks, hotspots, panels).
- **Object Localisation:** determining where these objects are located by drawing bounding boxes around them.

Modern object detection architectures are categorised into two families:

**Two-stage detectors** (e.g., R-CNN, Fast R-CNN, Faster R-CNN):

- Stage 1 generates region proposals using a Region Proposal Network.
- Stage 2 classifies and refines bounding box coordinates for each proposal.

The two-stage approach allows for high accuracy, as the second stage refines results, but it's generally slower because of the two-step process.

**One-stage detectors** (e.g., YOLO, SSD): predict bounding boxes and class labels directly from the full image in a single forward pass, enabling real-time inference while maintaining competitive accuracy.

## **1.8 Conclusion**

This chapter established the theoretical foundations necessary for the development of the proposed inspection system. The following chapter describes the system design and methodology adopted to implement the proposed solution.

---

## **Chapter 2**

*Methodology and System Design*

---

## Chapter 2

# Methodology and System Design

### 2.1 Introduction

This chapter describes the overall design of the proposed intelligent inspection system. It covers the dual-model strategy, model selection, image type classification model, data acquisition and preprocessing, dashboard architecture, and evaluation metrics used.

### 2.2 Dual-Model Strategy

The proposed system adopts a dual-model strategy based on image type. RGB and thermal images possess significantly different visual characteristics: RGB images contain texture, colour, and surface information, whereas thermal images represent temperature distributions and heat anomalies.

Because of these differences, using a single model for both modalities may reduce detection performance. To address this, two independent models were developed: one for visible-light (RGB) inspection and one for infrared thermography analysis.

### 2.3 Model Selection

The YOLO (You Only Look Once) object detection architecture was selected for photovoltaic fault detection in both RGB and thermal images for the following reasons:

- **Real-time detection capability:** high-speed inference suitable for near real-time analysis.
- **High detection accuracy:** strong performance in detecting small and localised defects such as hotspots, cracks, and bypass diode anomalies.
- **Efficient localisation:** predicts bounding boxes with confidence scores, enabling precise defect localisation.
- **Computational efficiency:** good balance between accuracy and computational cost compared to heavier architectures.
- **Scalability:** capable of efficiently processing large image datasets acquired during UAV

inspection.

Two independent models were developed:

- YOLOv8 was chosen for infrared thermography analysis for its proven stability, efficiency, and good performance in detecting high-contrast heat-based anomalies while maintaining low computational cost. This was particularly suitable given the limited GPU resources during training.
- YOLOv11 was chosen for visible-light inspection because RGB defect detection relies more on fine-grained visual details such as cracks, stains, and small surface variations. The improved feature extraction capabilities of YOLOv11 make it more suitable for capturing these subtle textures and enhancing detection accuracy in complex visible-light environments.

Therefore, the choice of models was influenced by both: the nature of the data (thermal vs RGB), and practical hardware constraints during experimentation.

## 2.4 Image Type Classification Model

To ensure that each image is processed by the appropriate detection model, a deep learning-based binary classification model based on a modified **ResNet18** architecture is used to distinguish between RGB and thermal images.

The model is trained using a custom dataset designed to distinguish between RGB and thermal photovoltaic images.

The dataset is organized into two main subsets: training set and validation set. Each subset contains two class-specific folders:

rgb/ containing visible-spectrum photovoltaic images

thermal/ containing infrared photovoltaic images.

## 2.5 Data Acquisition For Detection Models

### 2.5.1 Thermal Dataset

The thermal dataset was obtained from the Kaggle platform [1] and contains infrared images of photovoltaic modules presenting different electrical anomalies and thermal defects.

The dataset includes a total of 7 339 thermal images annotated in YOLO format, organised into 3 subsets:

- training
- validation

- test

Each subset contains 2 folders:

- images
- labels

In addition, a YAML configuration file is provided to define the dataset structure and class names required for YOLO training.

The eight defect classes are listed in Table 2.1.

**Table 2.1.** *Thermal dataset classes*

| Class ID | Defect Class           |
|----------|------------------------|
| 0        | MultiByPassed          |
| 1        | MultiDiode             |
| 2        | MultiHotSpot           |
| 3        | SingleByPassed         |
| 4        | SingleDiode            |
| 5        | SingleHotSpot          |
| 6        | StringOpenCircuit      |
| 7        | StringReversedPolarity |

**Preprocessing and augmentation:** Each image was preprocessed as follows:

- auto-orientation of pixel data with EXIF removal
- resizing to  $640 \times 640$  pixels
- grayscale conversion

Augmentation techniques applied to improve model robustness and increase dataset variability include:

- horizontal and vertical flipping
- 90-degree rotations
- random shear transformations and brightness adjustments

**annotation information:** All anomalies in the dataset are annotated using bounding boxes and class labels compatible with the YOLO object detection format. Each annotation file contains: the class identifier, normalized bounding box center coordinates, normalized width and height values.

The YOLO annotation format is represented as follows:

```
7 0.5421875 0.52890625 0.8171875 0.2625
```

**Figure 2.1.** *Example of YOLO annotation format*

- **Class ID:** 7 → StringReversedPolarity
- **Normalized bounding box center coordinates:** 0.5421875 → the center of the bounding box is located at 54.22% of the image width measured from the left side of the image.  
0.52890625 → the center of the bounding box is located at 52.89% of the image height measured from the top of the image.
- **Normalized bounding box dimensions:** 0.8171875 → the bounding box width occupies approximately 81.72% of the image width.  
0.2625 → the bounding box height occupies approximately 26.25% of the image height.

### 2.5.2 RGB Dataset

The RGB dataset was obtained from the Roboflow Universe platform [2] and contains visible-spectrum images of photovoltaic panels presenting different surface defects. The dataset contains 6 493 RGB images annotated in YOLO format, organised into training, validation, and test subsets. Each subset is split into two folders: images and labels. A yaml configuration file is also provided to define the dataset structure and class names required for YOLO training.

The classes are listed in Table 2.2.

**Table 2.2.** *RGB dataset classes*

| Class ID | Defect Class |
|----------|--------------|
| 0        | bird_drop    |
| 1        | cracked      |
| 2        | dusty        |
| 3        | panel        |

**Preprocessing and augmentation:** Each image was preprocessed as follows:

- auto-orientation of pixel data with EXIF removal
- resizing to  $640 \times 640$  pixels

No data augmentation was applied to the RGB dataset.

**annotation information:** Similar to the thermal dataset, the RGB dataset annotations follow the YOLO object detection format.

## 2.6 Dashboard Design

### 2.6.1 System Architecture

The proposed dashboard architecture consists of three main layers:

**Input layer:** The dashboard receives photovoltaic images uploaded by the user, either RGB or thermal. Once an image is uploaded, it is transmitted to the processing pipeline.

**Processing layer:**

- The system first determines whether the uploaded image is RGB or thermal using the classifier
- Based on the predicted image type, the corresponding YOLO model is selected and performs defect detection, outputting defect class labels, bounding boxes, and confidence scores.

**Output layer:** The dashboard displays the annotated image containing bounding boxes, detected fault names, confidence scores, in addition to severity levels, maintenance recommendations, total number of faults, number of urgent faults and the average confidence.

### 2.6.2 Fault Severity Classification

Each detected defect is associated with a predefined severity level to facilitate maintenance prioritisation. Severity levels are determined according to operational impact, risk of power loss, potential safety hazards, and probability of defect propagation.

**Table 2.3.** *Defect severity classification for thermal and RGB models*

| Defect Type            | Severity Level |
|------------------------|----------------|
| MultiByPassed          | Medium         |
| MultiDiode             | High           |
| MultiHotSpot           | High           |
| SingleByPassed         | Low            |
| SingleDiode            | Medium         |
| SingleHotSpot          | Medium         |
| StringOpenCircuit      | Critical       |
| StringReversedPolarity | Critical       |
| bird_drop              | Low            |
| cracked                | High           |
| dusty                  | Low            |

Urgent faults refer to defects classified as either High or Critical severity, as both levels require

immediate intervention to prevent further performance degradation or system damage.

### 2.6.3 Maintenance Recommendations

The system integrates a rule-based layer that transforms model predictions into actionable maintenance decisions.

Table 2.4 presents the maintenance actions for thermal defects, and Table 2.5 for RGB defects.

**Table 2.4.** *Maintenance actions for thermal defect classes*

| Defect Type            | Maintenance Action                                                   |
|------------------------|----------------------------------------------------------------------|
| MultiByPassed          | Inspect and replace faulty bypass diodes, check affected modules     |
| MultiDiode             | Urgent inspection and replacement of defective diodes                |
| MultiHotSpot           | Immediate intervention, inspect cells and replace damaged modules    |
| SingleByPassed         | Monitor panel, schedule maintenance if condition worsens             |
| SingleDiode            | Inspect diode and plan replacement to avoid efficiency loss          |
| SingleHotSpot          | Inspect affected cell, clean panel and monitor temperature evolution |
| StringOpenCircuit      | Immediate repair required, check wiring and reconnect the string     |
| StringReversedPolarity | Immediate shutdown and correct wiring to prevent system damage       |

**Table 2.5.** *Maintenance actions for RGB defect classes*

| Defect Type | Maintenance Action                                                   |
|-------------|----------------------------------------------------------------------|
| bird_drop   | Clean the panel surface to restore efficiency                        |
| cracked     | Replace the damaged panel to avoid performance loss and safety risks |
| dusty       | Perform routine cleaning to improve energy output                    |

## 2.7 Evaluation Metrics

### 2.7.1 Intersection over Union

In object detection tasks, two bounding boxes are involved:

- Predicted bounding box: The bounding box generated by the detection model.

- Ground-truth bounding box: The manually annotated bounding box provided in the dataset labels during dataset preparation.

Intersection over Union (IoU) quantifies the degree of spatial overlap between a predicted bounding box and the ground-truth bounding box:

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (2.1)$$

### 2.7.2 Precision, Recall, and F1-Score

Before delving into metrics, let's establish some fundamental concepts:

**True Positive (TP):** the model correctly detects an object with IoU above the threshold.

**False Positive (FP):** the model detects an object that does not exist, or IoU is below the threshold.

**False Negative (FN):** the model fails to detect an existing object.

|                  |              | Actual Values |              |
|------------------|--------------|---------------|--------------|
|                  |              | Positive (1)  | Negative (0) |
| Predicted Values | Positive (1) | TP            | FP           |
|                  | Negative (0) | FN            | TN           |

**Figure 2.2.** Illustration of True Positive, False Positive, and False Negative detection cases.

#### Precision:

Is a critical metric in model evaluation as it serves to quantify the accuracy of the positive predictions made by the model.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.2)$$

#### Recall:

Recall measures the model's capability to capture all relevant objects in the image.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.3)$$

**F1-Score:**

Is the harmonic mean of precision and recall. It provides a balanced measure of the model's performance, considering both false positives and false negatives

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (2.4)$$

**2.7.3 Mean Average Precision**

Mean Average Precision (mAP) is computed by averaging the Average Precision (area under the Precision-Recall curve) across all defect classes.

- **mAP@0.5:** considers a detection correct if IoU  $\geq 0.50$ . It is a lenient metric.
- **mAP@0.5:0.95:** averages mAP at IoU thresholds from 0.50 to 0.95 in steps of 0.05. This rigorous metric rewards models that achieve high localization accuracy.

**2.7.4 Accuracy**

Accuracy is a standard evaluation metric used for classification tasks. It represents the proportion of all classifications that were correct, whether positive or negative.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.5)$$

**2.8 Conclusion**

This chapter presented the full system design, from the dual-model strategy and dataset preparation to the dashboard architecture and evaluation metrics. The next chapter details the technical implementation.

---

## **Chapter 3**

### *Implementation*

---

## Chapter 3

# Implementation

### 3.1 Introduction

This chapter details the technical realisation of the developed solution, covering the development environment, model training configurations, and the testing and validation workflow.

### 3.2 Development Environment

The implementation was carried out using the Python programming language. The development process was conducted in Visual Studio Code. Key libraries used include:

- **PyTorch:** primary deep learning framework for building and training the ResNet18 classifier.
- **Ultralytics YOLO:** used for implementing YOLOv8 and YOLOv11 object detection models.
- **Torchvision:** dataset handling and image transformations.
- **PIL (Python Imaging Library):** image preprocessing and visualisation.
- **Flask:** deploying trained models in a web-based dashboard interface.

### 3.3 Model Training

#### 3.3.1 Image Type Classifier (ResNet18)

A convolutional neural network based on ResNet18 was used for binary classification of input images into thermal or RGB categories.

All images were converted to grayscale to force the model to learn structural and texture-based differences rather than colour information.

The original ResNet18 pretrained on ImageNet was modified as follows:

1. Replacement of the first convolutional layer to accept single-channel (grayscale) input.
2. Modification of the final fully connected layer to output a single binary prediction.

Table 3.1 presents the training configuration for the classification model.

**Table 3.1.** *Image type classifier (ResNet18) – training configuration*

| Parameter     | Value                            |
|---------------|----------------------------------|
| Input size    | $224 \times 224$ pixels          |
| Loss function | Binary Cross-Entropy with Logits |
| Optimizer     | Adam                             |
| Learning rate | $1 \times 10^{-4}$               |
| Batch size    | 32                               |
| Epochs        | 20                               |

A weighted loss function was used to handle class imbalance between RGB and thermal samples.

### 3.3.2 Thermal Defect Detection Model (YOLOv8)

The thermal inspection model was implemented using the YOLOv8 framework by Ultralytics, chosen for its balance between speed and detection accuracy.

Table 3.2 presents the training configuration for the thermal defect detection model .

**Table 3.2.** *YOLOv8 thermal detection model – training configuration*

| Parameter  | Value                   |
|------------|-------------------------|
| Image size | $640 \times 640$ pixels |
| Batch size | 4                       |
| Epochs     | 50                      |

### 3.3.3 RGB Defect Detection Model (YOLOv11)

The RGB defect detection model is based on YOLOv11, trained separately using a dataset of visible-spectrum solar panel images. This model focuses on visual surface defects.

Table 3.3 presents the training configuration for the RGB defect detection model .

**Table 3.3.** *YOLOv11 RGB detection model – training configuration*

| Parameter  | Value                   |
|------------|-------------------------|
| Image size | $640 \times 640$ pixels |
| Batch size | 8                       |
| Epochs     | 50                      |

### 3.4 Training Challenges

**GPU memory limitation:** Limited GPU memory constrained the YOLOv8 training batch size to 4, increasing training time and reducing convergence stability.

**Heterogeneity between thermal and RGB data:** Thermal images are intensity-based while RGB images are colour-based. Both types were converted to grayscale for the classifier to ensure consistent feature learning.

**Class imbalance:** The dataset used for the classifier training contained an unequal number of RGB and thermal samples. This was addressed using a weighted loss function (BCEWithLogitsLoss with `pos_weight`).

**Training time variability:** Training duration varies depending on GPU availability and system load.

### 3.5 Testing and Validation

After training, the final model was evaluated using a separate **test set** not used during training or validation. This ensures an unbiased evaluation of generalisation performance under real-world conditions.

The inference workflow is as follows:

1. Input image is uploaded through the Flask interface.
2. The classifier determines the image type (thermal or RGB).
3. Based on the prediction, the appropriate YOLO model is selected.
4. The selected model outputs bounding boxes, class labels, and confidence scores.
5. Results are rendered and returned to the user interface.

The object detection models were evaluated using standard object detection metrics: Precision, Recall, Mean Average Precision (mAP@50 and mAP@50-95).

The classification model was evaluated using validation accuracy.

### 3.6 Conclusion

This chapter detailed the technical implementation of the proposed system. The following chapter presents and discusses the experimental results obtained.

---

## **Chapter 4**

### *Results and Discussion*

---

## Chapter 4

# Results and Discussion

### 4.1 Introduction

This chapter presents the quantitative evaluation of the trained models and the qualitative analysis of inspection results through the developed dashboard interface.

### 4.2 Quantitative Evaluation

#### 4.2.1 Thermal Detection Model (YOLOv8)

##### Global performance

Table 4.1 presents the global performance metrics of the thermal detection model.

**Table 4.1.** *YOLOv8 thermal model – global performance metrics*

| Metric       | Value |
|--------------|-------|
| Precision    | 0.748 |
| Recall       | 0.717 |
| mAP@0.5      | 0.748 |
| mAP@0.5:0.95 | 0.577 |

- **Precision = 0.748** indicates that approximately 74.8% of detected bounding boxes correspond to real defects, while the remaining 25.2% are false positives.
- **Recall = 0.717** indicates that the model successfully identifies about 71.7% of all existing defects, with 28.3% being missed detections.
- **mAP@50= 0.748** shows good detection and classification performance with acceptable localisation accuracy at an IoU threshold of 0.5.
- **mAP@50-95= 0.577** reflects performance averaged across stricter IoU thresholds. The decrease relative to mAP@0.5 is expected and typical for complex real-world detection tasks.

### Per-class performance

The per-class metrics are detailed in Table 4.2.

**Table 4.2.** *YOLOv8 thermal model – per-class performance metrics*

| Class                  | Precision | Recall | mAP@50 | mAP@50-95 |
|------------------------|-----------|--------|--------|-----------|
| MultiByPassed          | 0.820     | 0.912  | 0.875  | 0.695     |
| MultiDiode             | 0.478     | 0.519  | 0.440  | 0.355     |
| MultiHotSpot           | 0.707     | 0.742  | 0.752  | 0.597     |
| SingleByPassed         | 0.888     | 0.905  | 0.929  | 0.758     |
| SingleDiode            | 0.795     | 0.636  | 0.727  | 0.571     |
| SingleHotSpot          | 0.807     | 0.737  | 0.828  | 0.648     |
| StringOpenCircuit      | 0.817     | 0.725  | 0.827  | 0.625     |
| StringReversedPolarity | 0.672     | 0.562  | 0.610  | 0.371     |

The model demonstrates **strong performance** on several defect classes:

- **SingleByPassed** achieves the highest performance (P = 0.888, R = 0.905, mAP@50 = 0.929, mAP@50-95 = 0.758), indicating that this defect type is well-learned and easily distinguishable.
- **SingleHotSpot**, **MultiHotSpot**, and **StringOpenCircuit** also show strong detection results, benefiting from clear and consistent thermal signatures.

Some classes remain more **challenging**:

- **MultiDiode** shows the weakest performance, indicating difficulty in distinguishing it from visually similar thermal patterns.
- **StringReversedPolarity** shows relatively lower performance, likely due to limited training samples and higher variability in thermal appearance.

#### 4.2.2 RGB Detection Model (YOLOv11)

##### Global performance

Table 4.3 presents the global performance metrics of the RGB detection model.

**Table 4.3.** *YOLOv11 RGB model – global performance metrics*

| Metric    | Value |
|-----------|-------|
| Precision | 0.864 |
| Recall    | 0.802 |
| mAP@50    | 0.877 |
| mAP@50-95 | 0.720 |

- **Precision = 0.864** means that approximately 86.4% of the detected bounding boxes correspond to real RGB defects or photovoltaic components, while the remaining predictions are false positives.
- **Recall = 0.802** indicates that the model successfully detects about 80.2% of all existing objects and defects present in the dataset, while the remaining detections are missed.
- **mAP@50 = 0.877** indicates very good detection and localization performance for RGB photovoltaic inspection.
- **mAP@50-95 = 0.720** A value of 0.720 shows that the RGB model maintains strong localization accuracy even under more demanding evaluation conditions.

### Per-class performance

The per-class metrics are detailed in Table 4.4.

**Table 4.4.** *YOLOv11 RGB model – per-class performance metrics*

| Class     | Precision | Recall | mAP@50 | mAP@50-95 |
|-----------|-----------|--------|--------|-----------|
| Bird_drop | 0.752     | 0.625  | 0.683  | 0.463     |
| Cracked   | 0.915     | 0.737  | 0.909  | 0.703     |
| Dusty     | 0.827     | 0.926  | 0.948  | 0.828     |
| Panel     | 0.962     | 0.919  | 0.966  | 0.884     |

Overall, the RGB detection model demonstrates strong performance across classes, with high precision and mAP values indicating accurate object localization and classification capabilities.

The model shows especially strong performance on several classes, particularly:

- **panel** has highest overall performance (**P = 0.962, R = 0.919, mAP@50 = 0.966, mAP@50-95 = 0.884**), indicating that photovoltaic panels are very easily distinguishable.
- **dusty** also achieved excellent results showing that dust accumulation patterns are effectively recognized by the model.
- **cracked** defects also produced strong detection performance demonstrating the capability of the model to detect visible structural damage on photovoltaic panels.

However, **bird-drop** achieved slightly lower performance compared to the other RGB classes,

particularly in recall and mAP@50-95 values.

Nevertheless, the obtained results remain satisfactory overall.

The slightly lower scores may be explained by the limited number of samples available in the dataset and the variability in the appearance, size, and shape of bird droppings on photovoltaic panels.

### 4.2.3 Image Type Classifier (ResNet18)

**Table 4.5.** *Image type classifier – validation performance*

| Metric              | Value  |
|---------------------|--------|
| Validation Accuracy | 99.92% |

The classifier achieves a validation accuracy of 99.92%, demonstrating its ability to reliably distinguish between thermal and RGB images before routing to the appropriate detection model.

## 4.3 Qualitative Results and Visual Analysis

### 4.3.1 Interface Overview

The developed interface offers two inspection modes, selectable at the top of the dashboard: Image Inspection and Video Inspection.

#### Image Inspection Mode:

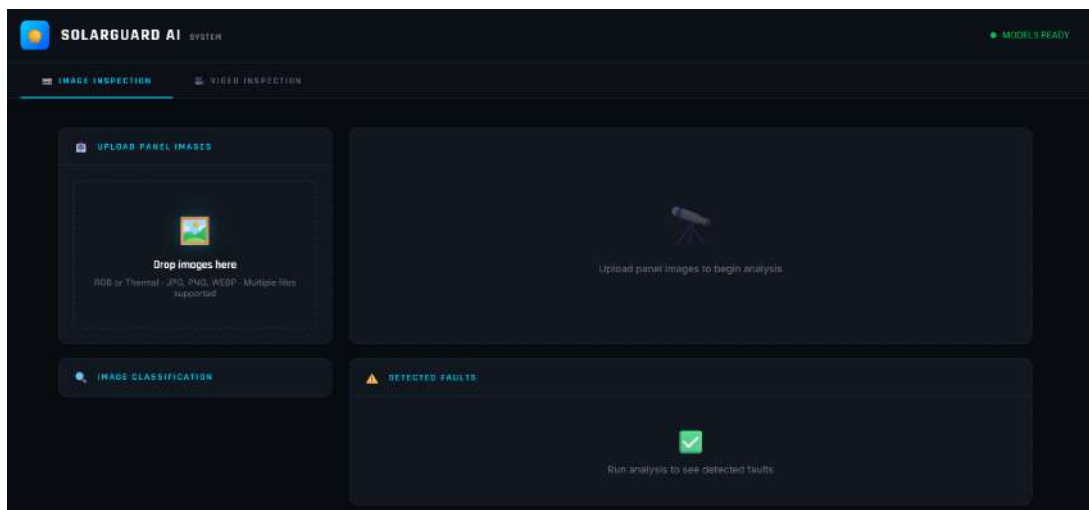
- **Image upload section:** Located in the top-left area of the interface and labeled “Drop image here”. It allows the user to upload one or several thermal or RGB images of photovoltaic panels at once, enabling efficient inspection of multiple panels in a single session.
- **Analyze Panel button:** Positioned below the upload section, and only appears after uploading an image. It launches the automatic inspection and defect detection process.
- **Classification result area:** Displays the predicted image type (thermal or RGB), together with the associated classifier confidence score after image analysis.
- **Detection visualization area:** Located on the right side of the interface. It displays the uploaded image annotated with bounding boxes, detected fault labels and confidence scores generated by the YOLO models.
- **Detected Faults section:** Positioned below the annotated image. For each detected anomaly, the system provides: defect class, confidence score, severity level and a recommended action.
- **Inspection summary indicators:** Displayed dynamically after analysis and include: total number of detected faults, number of urgent faults and average confidence score of

detections.

### Video Inspection Mode:

In this mode, the user uploads a video file captured during a drone flyover of the photovoltaic installation. The system processes the video by applying the detection pipeline frame by frame, enabling continuous and automated inspection of large solar farms without requiring manual image extraction. This mode is particularly suited for real-world deployment scenarios where drone footage is directly fed into the system.

The interface not only visualizes defects but also assists maintenance decision-making.



**Figure 4.1.** Dashboard interface prior to image upload.

### 4.3.2 Thermal Inspection Results

In this session, a batch of images containing both thermal and RGB panels was uploaded simultaneously, which are then listed in the left panel. Each uploaded image is displayed with its filename and a summary of the number of faults detected.

The user can navigate between the analyzed images using the Prev and Next navigation buttons located at the top of the detection area.

Figure 4.2 shows an example of thermal photovoltaic panel inspection performed using the developed system.

The classifier automatically determined the type of the selected image with a confidence of 100%, allowing the system to automatically select the YOLOv8 thermal model.

Two anomalies were detected and correctly localized using bounding boxes drawn around the abnormal thermal regions, with the display of the defect class name and the confidence score.

- one **SingleHotSpot** defect, classified as *Medium* severity,
- one **MultiHotSpot** defect, classified as *High* severity due to its larger impact on photovoltaic panel performance and potential risk of damage propagation.

Recommended actions were also displayed for each defect.

The interface also displayed the inspection summary indicators, including:

- total number of detected faults: 2,
- number of urgent faults: 1,
- average confidence score: 60%.

At the top of the detection view, a fault highlight bar displays the most important fault detected in the currently viewed image.

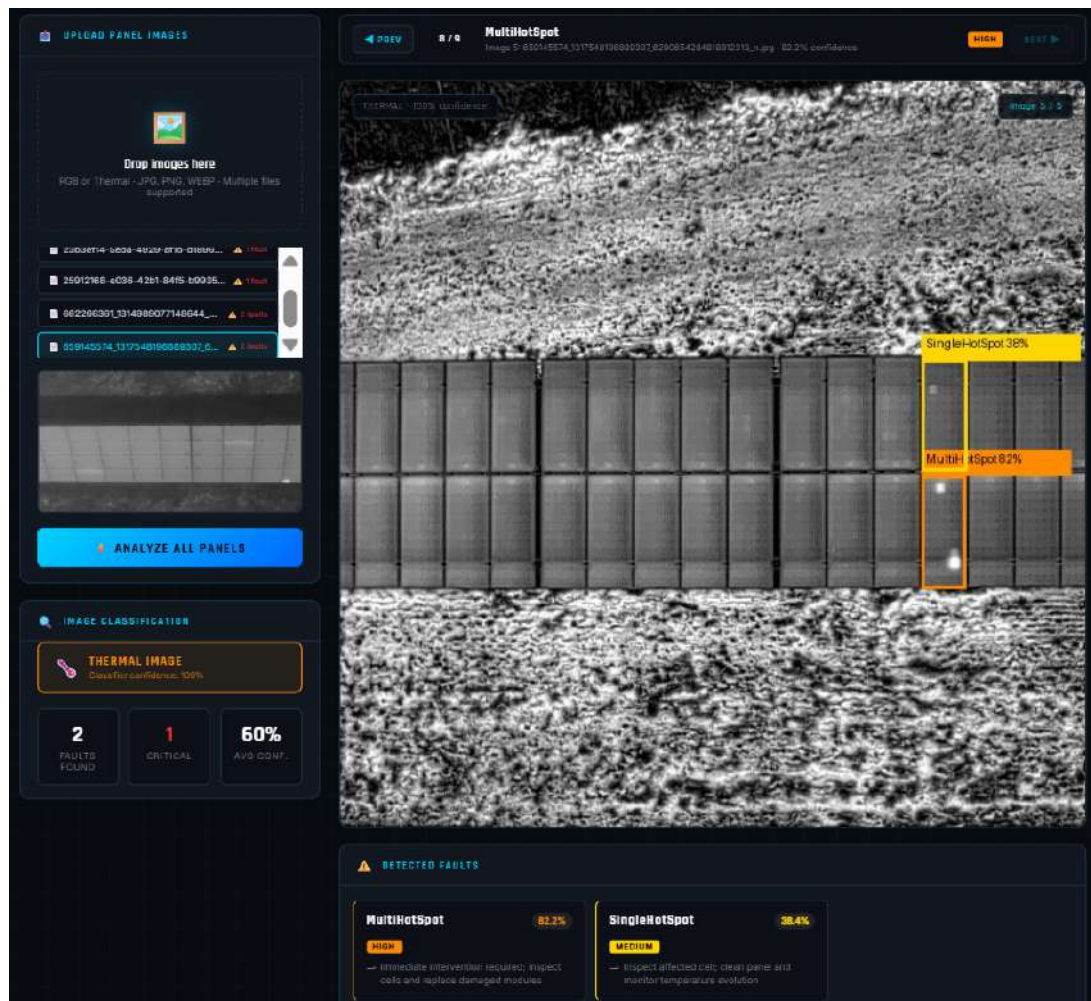


Figure 4.2. Dashboard result for thermal image inspection.

### 4.3.3 RGB Inspection Results

Figure 4.3 presents an example of RGB photovoltaic panel inspection performed using the developed system.

The uploaded image was correctly classified as an RGB image with a classifier confidence of 100%, allowing the system to automatically select the YOLOv11 RGB detection model.

The model successfully detected two surface defects of type **cracked**.

In addition, the photovoltaic panel itself was also identified through the **panel** class, providing an additional localisation reference.

Both crack defects were correctly localised using bounding boxes drawn around the damaged regions of the panel.

For each detected anomaly, the system displayed the defect class name, the corresponding confidence score, the severity level, and the associated maintenance recommendation.

Since cracked panels can significantly affect photovoltaic performance and may present safety risks through water ingress and electrical hazards, both detected defects were classified as **High** severity anomalies requiring corrective maintenance actions.

The interface also displayed the inspection summary indicators, including:

- total number of detected faults: 2,
- number of urgent faults: 2,
- average confidence score: 55%.

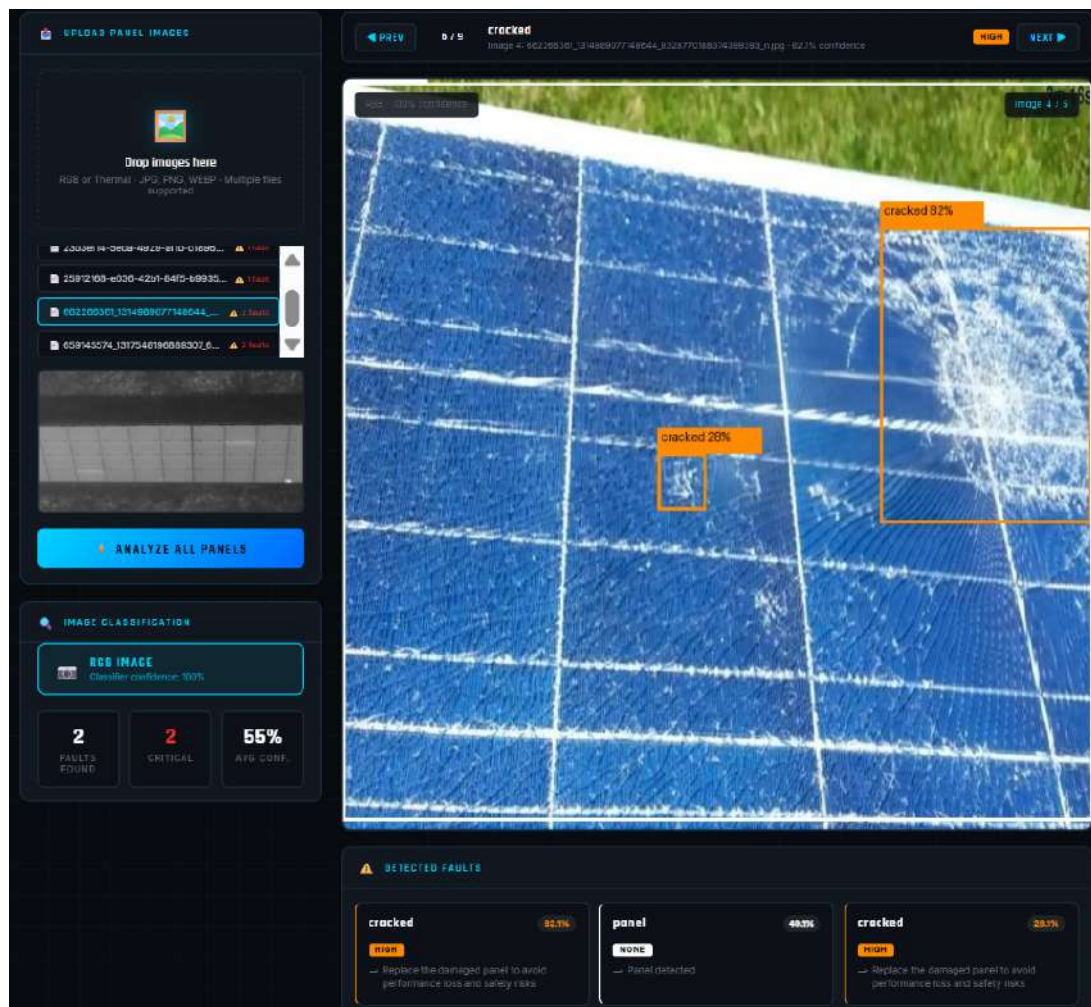


Figure 4.3. Dashboard result for RGB image inspection.

## 4.4 Performance Analysis

### 4.4.1 Strengths of the Proposed System

- YOLO-based detection enables fast and accurate real-time defect localisation for both thermal and RGB modalities.
- The thermal YOLOv8 model achieves a global mAP@0.5 of 74.8%, with particularly strong performance on SingleByPassed (92.9%), SingleHotSpot (82.8%), and StringOpenCircuit (82.7%).
- The RGB YOLOv11 nano model achieves a global mAP@0.5 of 87.7%, with excellent results on Panel (96.6%), Dusty (94.8%), and Cracked (90.9%) classes.
- The automatic image-type classifier (ResNet18) achieves a validation accuracy of 99.92%, ensuring near-perfect routing of input images to the appropriate detection model.
- The dashboard integrates defect localisation, confidence scores, severity classification, and maintenance recommendations into a practical and interpretable decision-support tool.

### 4.4.2 Weaknesses of the Proposed System

- Certain thermal defect classes (MultiDiode, StringReversedPolarity) are more difficult to distinguish due to visual similarity, limited dataset representation, and variability in thermal appearance.
- RGB detection performance is sensitive to image quality, lighting conditions, and occlusions.
- Very small defects such as micro-cracks may not always be clearly visible, leading to missed detections or reduced confidence scores.

## 4.5 Limitations

**Dataset limitations:** Some defect classes contain fewer training samples than others, which may negatively affect model generalisation for rare anomalies.

Datasets collected from different sources introduce variability in image quality and acquisition conditions.

**Hardware limitations:** Limited GPU memory constrained training configurations, increasing duration and limiting experimentation with larger architectures.

**Annotation quality:** Detection performance depends on annotation precision. Imperfect bounding boxes and visually ambiguous regions may affect localisation accuracy and model learning quality.

## **4.6 Conclusion**

The developed system demonstrated good overall performance in both thermal and RGB defect detection tasks. The integration of the image-type classifier, dual detection models, and a structured dashboard transforms the system into a practical maintenance decision-support tool.

---

## **Chapter 5**

*Conclusion and Perspectives*

---

## Chapter 5

# Conclusion and Perspectives

### 5.1 Conclusion

This project focused on the design and implementation of an intelligent photovoltaic inspection system capable of detecting and classifying defects using both thermal and RGB images. The proposed system integrates YOLOv8 and 11 deep learning models together with a ResNet18 image-type classifier and a Flask-based decision support interface.

The system operates as a two-stage pipeline: an image classifier first identifies the input modality (thermal or RGB), then a specialised YOLO model performs defect detection. The system successfully detects multiple types of photovoltaic anomalies, including thermal faults such as hotspots and diode-related issues, and visible defects such as cracks, dust accumulation, and bird droppings. The interface provides structured outputs including bounding boxes, confidence scores, severity levels, and maintenance recommendations.

The project demonstrates that combining thermal and RGB analysis significantly improves the completeness of photovoltaic inspection, enabling both internal electrical faults and external surface defects to be identified effectively. This solution reduces the need for manual inspections, which are often time-consuming, costly, and potentially risky for technicians working in large solar farms. Early detection of faults helps prevent energy losses and extends the operational life-time of photovoltaic systems. On a broader scale, improving the efficiency and reliability of solar energy systems contributes to increased renewable energy adoption, supporting environmental sustainability.

### 5.2 Future Work

Several directions can be pursued to further enhance the system:

- **Real-time drone integration:** Embedding the inspection pipeline into operational drone platforms for automated large-scale PV monitoring.
- **Dataset expansion:** Increasing the number and diversity of training samples, especially for underrepresented defect classes such as MultiDiode and StringReversedPolarity.

- **Higher-performance computational resources:** Training on more powerful GPUs to allow larger batch sizes, higher image resolutions, and more advanced model architectures.
- **Edge deployment:** Optimising models for embedded devices to allow real-time inference directly on drones.
- **Multimodal fusion:** Exploring advanced fusion strategies between thermal and RGB data to further improve detection reliability in complex scenarios.

# Bibliography

- [1] Darabi, P. *Solar Panel Thermal Anomalies Detection Dataset*. Kaggle, 2024. Available at: <https://www.kaggle.com/datasets/pkdarabi/solarpanel>
- [2] Susan (Roboflow Universe). *Panel Solar BW945 Dataset*. 2024. Available at: <https://universe.roboflow.com/susan-ifblr/panel-solar-bw945/dataset/1>
- [3] Sitemark. *Troubleshooting Lesser-Known Solar Module Anomalies*. Available at: <https://www.sitemark.com/blog/troubleshooting-lesser-known-solar-module-anomalies>
- [4] Ashhab, M. S. et al. *PV Module Fault Detection Using AI Techniques*. International Journal of Image, Graphics and Signal Processing, vol. 10, no. 6, 2018. Available at: <https://www.mecs-press.org/ijigsp/ijigsp-v10-n6/IJIGSP-V10-N6-2.pdf>
- [5] Wavetra. *How to Read Solar Panel PV Module Technical Data Sheet*. Available at: <https://wavetra.com/blogs/news/how-to-read-solar-panel-pv-module-technical-data-sheet/>
- [6] Photovoltaikbüro. *PV Inspection and Analysis Services*. Available at: <https://photovoltaikbuero.de/en/services/>
- [7] Masson, G. et al. *Trends in Photovoltaic Applications 2025*. IEA Photovoltaic Power Systems Programme (IEA-PVPS), 2025. Available at: [https://iea-pvps.org/trends\\_reports/trends-2025/](https://iea-pvps.org/trends_reports/trends-2025/)
- [8] PVEducation. *Module Circuit Design*. 2019. Available at: <https://www.pveducation.org/pvcdrom/modules-and-arrays/module-circuit-design>